d) ssl-client-subject-dn

e) ssl-client-naughty-or-nice

11.4.4 Using a service mesh

Although TLS certificate authentication is very secure, client certificates still must be generated and distributed to clients, and periodically renewed when they expire. If the private key associated with a certificate might be compromised, then you also need to have processes for handling revocation or use short-lived certificates. These are the same problems discussed in chapter 10 for server certificates, which is one of the reasons that you installed a service mesh to automate handling of TLS configuration within the network in section 10.3.2.

To support network authorization policies, most service mesh implementations already implement mutual TLS and distribute both server and client certificates to the service mesh proxies. Whenever an API request is made between a client and a server within the service mesh, that request is transparently upgraded to mutual TLS by the proxies and both ends authenticate to each other with TLS certificates. This raises the possibility of using the service mesh to authenticate service clients to the API itself. For this to work, the service mesh proxy would need to forward the client certificate details from the sidecar proxy to the underlying service as a HTTP header, just like you've configured the ingress controller to do. Istio supports this by default since the 1.1.0 release, using the `X-Forwarded-Client-Cert` header, but Linkerd currently doesn't have this feature.